Mohammad Ayaz Alam

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SKILLS

Generative AI Systems:	Agentic AI, LangGraph, LangChain, LangSmith, MCP, RAG
Natural Language Processing:	Hugging Face, Transformers, Fine-tuning, Prompt Engineering
Machine Learning:	PyTorch, Scikit-learn, XGBoost, LightGBM, Feature Engineering, Model Evaluation
Programming:	Python, SQL, JavaScript
Backend & Web Development:	FastAPI, Django, Flask, Next.js, REST APIs
Cloud & DevOps:	AWS (EC2, S3, Lightsail), Docker, Git/GitHub, CI/CD, Linux
Databases:	PostgreSQL, Firebase, Query Optimization
Data & Visualization:	Pandas, NumPy, Power BI, Plotly
AI-Assisted Development:	Claude Code, OpenAI Codex, Cursor, GitHub Copilot
Other:	Flutter, HTML, CSS, LaTeX

EXPERIENCE

Zalion GmbH, *Junior AI Engineer* | Munich, Germany — Mar 2026 – Jul 2026

- Developed multi-agent procurement AI workflows, enabling specialized agents to execute task-specific operations.
- Integrated AI agent endpoints into conversational interfaces for interactive workflow execution.
- Built REST APIs with FastAPI and frontend components with Next.js for AI-powered applications.
- Optimized agent orchestration and execution workflows to reduce unnecessary processing and improve responsiveness.
- Contributed to AWS EC2 deployments, Docker infrastructure, and CI/CD pipelines.

Siemens Energy, *Master's Thesis – Junior Data Scientist (NLP)* | Erlangen, Germany — Jul 2025 – Feb 2026

- Designed a modular multi-agent framework using LangGraph and LangChain for industrial NLP applications.
- Implemented agent orchestration, tool execution, and persistent conversation state using PostgreSQL checkpointing.
- Developed FastAPI endpoints for agent and tool management, streaming responses, and OpenAI-compatible interfaces.
- Integrated execution audit trails and human-in-the-loop approval workflows for controlled external actions.
- Validated the framework across 2 industrial applications: Factory Assistance & Terminal Management System workflows.

Ilham Labs.ai, *Founder & Product Engineer* | Independent Product Initiative, Germany — 2025 – Present

- Owned architecture, full-stack development, deployment, and security across independent products.
- Built Lattice Lab, a visual multi-model AI workspace with parallel inference, branching workflows, and context merging.
- Implemented graph-based conversations, authenticated workspaces, and PostgreSQL persistence for Lattice Lab.
- Developed Air Lune, a multiplayer focus platform using FastAPI, Next.js, and LiveKit for real-time collaboration.
- Built WhenYouOpen.me, a scheduled message platform using FastAPI, AWS S3, Docker, and Nginx.

Institute for Employment Research (IAB), *Research Assistant* | Nuremberg, Germany — Nov 2024 – Feb 2026

- Applied XGBoost, Random Forest, SVM, and positive-unlabeled learning to labor-market analysis and job-offer ranking.
- Built scalable preprocessing and feature-engineering pipelines for datasets containing 90M+ records.
- Developed predictive models to rank job offers and support caseworker decision-making.
- Addressed class imbalance using weight estimation and alternative positive-unlabeled learning strategies.
- Conducted exploratory data analysis, model evaluation, visualization, and research reporting.

BASF, *Data Consultant Intern* | Ludwigshafen am Rhein, Germany — Apr 2025 – Jul 2025

- Developed Generative AI tools to support internal business workflows.
- Built a data-validation AI agent for structured-data checks, configurable tests, and quality scoring.
- Implemented automated validation reports to communicate data-quality issues and results.
- Developed an internal event application covering submissions, voting, analytics, and access controls.
- Contributed to internal digital tools for project management and operational workflows.

PROJECTS

Multi-Agent Research System | AI Research Automation - Link

- Built a LangChain research pipeline for web search, content extraction, report generation, and review.
- Implemented specialized search and reader agents, followed by writer and critic chains.
- Integrated Tavily and Beautiful Soup for web retrieval, with LLM-based report scoring.
- Deployed a Streamlit app with structured reports, source citations, and Markdown exports.

Retail Sales Forecasting (M5 Dataset) | Time-Series Machine Learning - Link

- Built a retail demand forecasting pipeline covering 30,490 product-store time series with a 28-day forecast horizon.
- Analyzed historical sales, pricing, and calendar data to identify demand trends and seasonal patterns.
- Engineered lag, rolling, and calendar features with safeguards against future-data leakage.
- Benchmarked ARIMA, SARIMA, LightGBM, and seasonal-naive models using chronological validation, achieving 0.7595 mean RMSSE with LightGBM, a 23.8% reduction against the weekly seasonal-naive baseline.

Fine-Tuned GPT-2 for Poetry Generation with Reinforcement Learning - Link

- Fine-tuned GPT-2 (124M) on a poetry corpus using Hugging Face Transformers.
- Designed custom reward functions for rhyme, coherence, and creativity.
- Applied policy-gradient reinforcement learning to optimize poetry generation.
- Developed a training pipeline combining supervised fine-tuning and reward-based optimization.

CERTIFICATES

Huggingface - AI Agents Fundamentals - Agents fundamentals, Tools and Action, Agents Workflow
iNeuron.ai - Full Stack Data Analytics - Python, SQL, Snowflake, PowerBI, DAX, Excel
iNeuron.ai - Deep Learning Foundation - Neural Network, Backpropagation, CNN
Kaggle - Intro to Machine Learning - Model Validation, Underfitting-Overfitting, Random Forest
freeCodeCamp - Data Analysis with Python - Numpy, Pandas, Data Cleaning

EDUCATION

Friedrich-Alexander-Universität Erlangen-Nürnberg, *MSc. Data Science* | Erlangen, Germany — 2022 - Present
• Courses: Business Analytics, ML Time Series, ML in Finance, Mathematics of Learning, Linked Data

Guru Gobind Singh Indraprastha University, *Bachelors of Computer Applications* | Delhi, India — 2018 - 2021
• Courses: OOPs, Network Security, DBMS, Web Technologies, Data Warehousing and Data Mining

Languages

English	Professional proficiency
German	A2 proficiency
Hindi & Urdu	Native proficiency